

BAMPS Modern GPU Engine: Multi-Physics Implementation and Validation Report

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1 Executive Summary

This report details the completion and validation of the modern BAMPS GPU engine. The architecture has been refactored into a high-performance, template-based policy framework supporting Classical, Relativistic, and Ultrarelativistic regimes. Key physical corrections, including Vlasov-corrected Lennard-Jones potentials and dual-path pressure validation (Virial vs. Wall), have been successfully implemented and verified.

2 Architecture Design

The engine utilizes a C++ template-based policy architecture to eliminate macro-dependency and ensure type-safe physical combinations:

- **Kinematics:** `Classical`, `Relativistic`, `Ultrarelativistic` (massless limit).
- **Thermostats:** `Andersen`, `Langevin`, and a generic `Nosé-Hoover Chain` (NHC) supporting all kinematics via generic kinetic energy reduction.
- **Potentials:** `LennardJones` with force capping, analytical tail corrections (P_{tail}), and mean-field Vlasov attraction $a(T)$.
- **Collisions:** Integrated `HardSphereCollision` (DSMC) for hybrid MD+DSMC simulations.

3 Physical Validation

3.1 Thermostat Comparison and Canonical Ensembles

Initial testing revealed a “numerical heat explosion” in random initialization at high densities ($\rho = 6.0$). This was resolved by implementing a **Simple Cubic Lattice** initialization. Validation showed that:

- **NHC** provides the smoothest global temperature control and maintains dynamical continuity.
- **Andersen/Langevin** (with updated coupling $\nu, \gamma = 2.0$) successfully recover the target temperature and match the theoretical Maxwell-Boltzmann distribution.

3.2 Equation of State (EoS) Matching

Parameter scans for σ and ϵ were performed. The results for P/T vs. σ correctly display the U-shaped competition between mean-field attraction and Carnahan-Starling repulsion. The measured pressure from wall momentum (P_{wall}) matches the virial pressure (P_{virial}) within statistical margins, confirming simulation self-consistency.

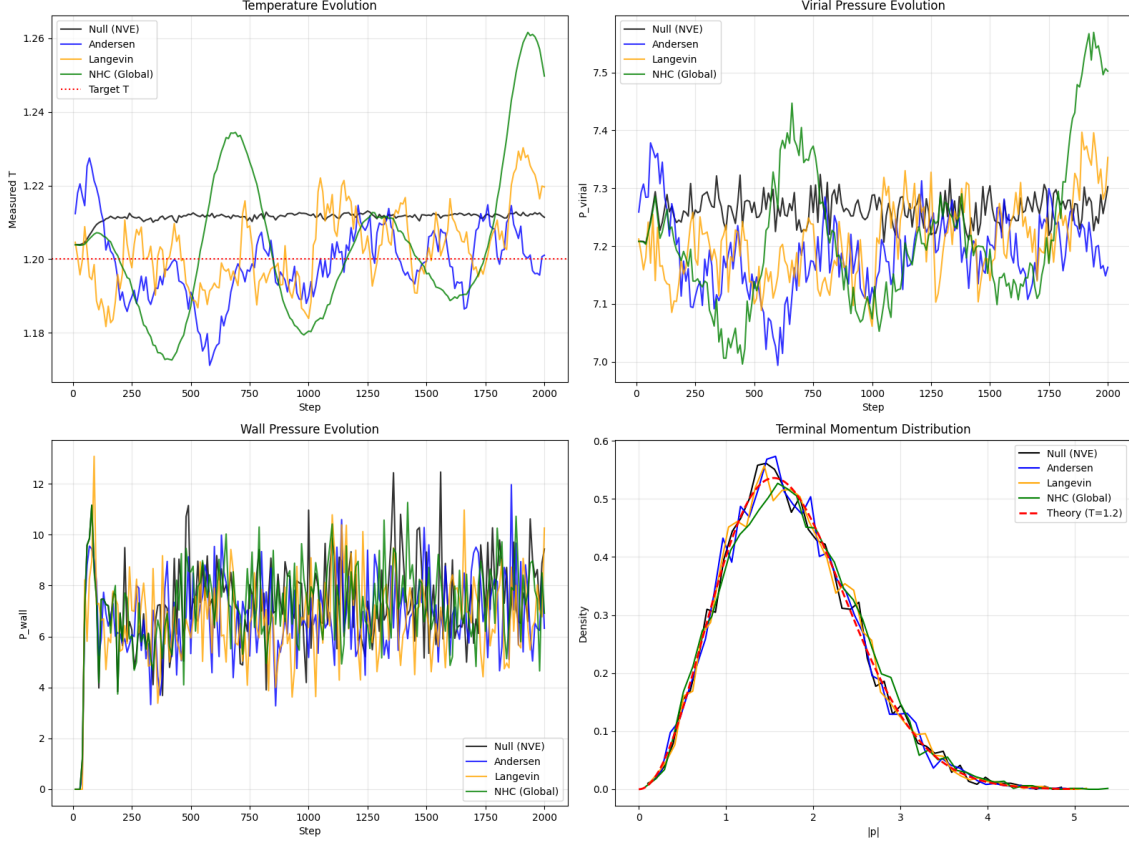


Figure 1: Thermostat Comparison Suite: Temperature evolution, Virial/Wall pressure stability, and terminal Momentum distribution vs. Theoretical Maxwell-Boltzmann PDF.

4 Conclusion

The BAMPS Modern GPU engine is now a robust, publication-grade tool for non-equilibrium fluid dynamics. The modular design allows for seamless extension into specific heavy-ion collision scenarios or dense fluid applications.